

Meeting 3 - 24/07/2026

Key Discussion Points

Energy Generation

Two solar generation models were considered:

- **Simple linear/angle-based model:**

$$P = P_{\text{max}} \times \cos(\gamma)$$

- **Detailed hardware-based model:** incorporates solar panel area, cell efficiency, solar constant, system efficiency, panel angle, and reflectivity.

Decision: Go with detailed one

OpenSN Extension Points

OpenSN can be extended through:

1. **Container images** — behaviour and applications running inside satellite containers.
2. **Topology Controller** — controls satellite positions, links, and dynamic configuration.
3. **Static configuration** — stores satellite-specific parameters.

The proposed energy model can use these existing extension mechanisms without modifying OpenSN's core networking engine.

Energy Consumption

Three potential consumption models were considered (Check slides for details):

- RF/EIRP-based model — requires antenna/RF parameters that OpenSN does not currently provide.
- DVFS-based model — requires physical CPU frequency information that is not available from the container.
- **Linear CPU-utilisation model** — uses container CPU utilisation as a proxy for satellite computing activity.

Decision: Use the **linear CPU-utilisation model**:

$$P_{\text{consume}} = P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) \times u(t)$$

Here, $u(t)$ represents CPU utilisation, while P_{idle} and P_{max} are satellite-specific assumed specifications.

Important limitation: CPU utilisation represents **relative computational activity**, not actual physical satellite power consumption.

Proposed Energy Architecture : Hybrid (Central + Local)

Central — Topology Controller

Responsible for:

- Satellite position
- Sun angle
- Eclipse detection
- Solar power generation (P_{gen})
- Passing generation information to each satellite

Local — Satellite Container

Responsible for:

- Reading its own CPU utilisation
- Calculating power consumption
- Updating its battery state
- Maintaining its local energy state

This keeps orbital calculations centralised while allowing each satellite to monitor its own computational activity.
